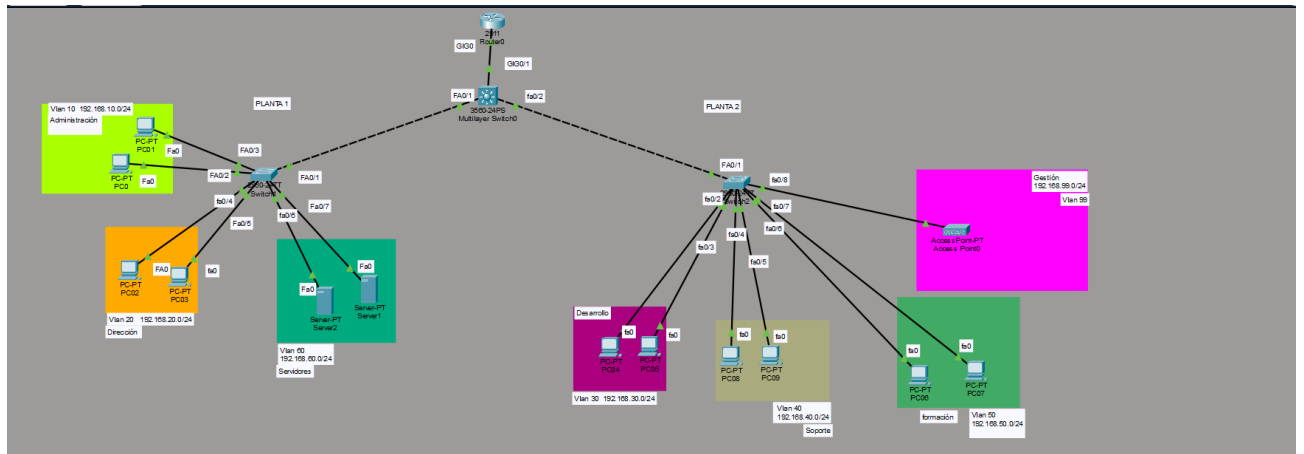


Proyecto intermodular de Administración y planificación de redes.

1. En primer lugar, creo la planificación de la infraestructura de red en packet tracer.

Decido usar un switch de capa 3 por su facilidad de gestionamiento.



2. Creo las Vlan en el switch de capa 3.

```
Multilayer Switch0
Physical Config CLI Attributes
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name DIR
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name DEV
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name SOPORTE
Switch(config-vlan)#vlan 50
Switch(config-vlan)#name FORMACION
Switch(config-vlan)#vlan 60
Switch(config-vlan)#name SERVIDORES
Switch(config-vlan)#valn 99
^
% Invalid input detected at '^' marker.
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name MGMT
Switch(config-vlan)#
```

3. Creo las interfaces VLAN en el mismo switch de capa 3

```

Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 30
Switch(config-if)# ip address 192.168.30.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 40
Switch(config-if)# ip address 192.168.40.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 50
Switch(config-if)# ip address 192.168.50.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 60
Switch(config-if)# ip address 192.168.60.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 99
Switch(config-if)# ip address 192.168.99.1 255.255.255.0
Switch(config-if)# no shutdown

```

4. Configuro los enlaces trunk .

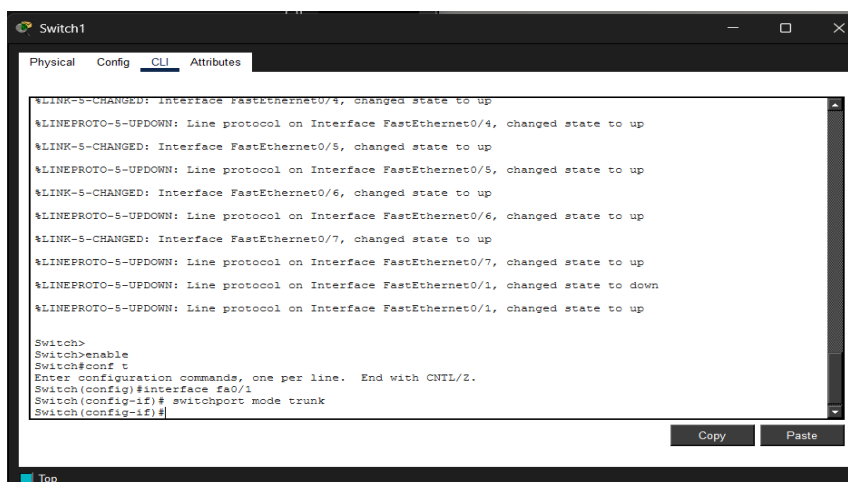
Primero en el switch de capa 3

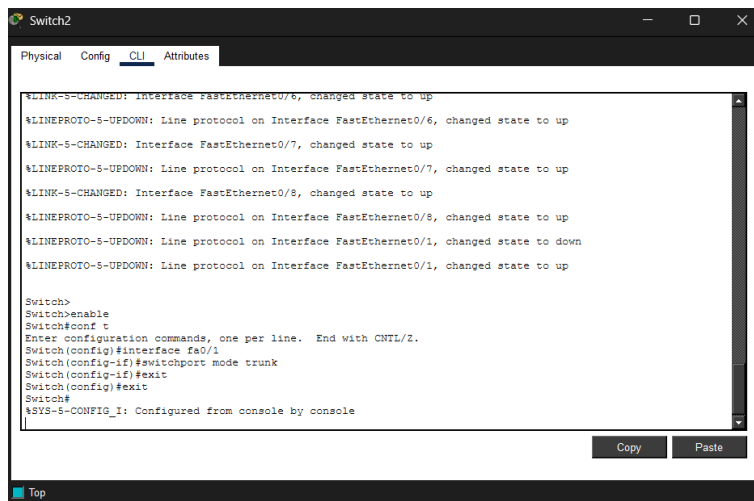
```

Switch#
Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)# switchport mode trunk

```

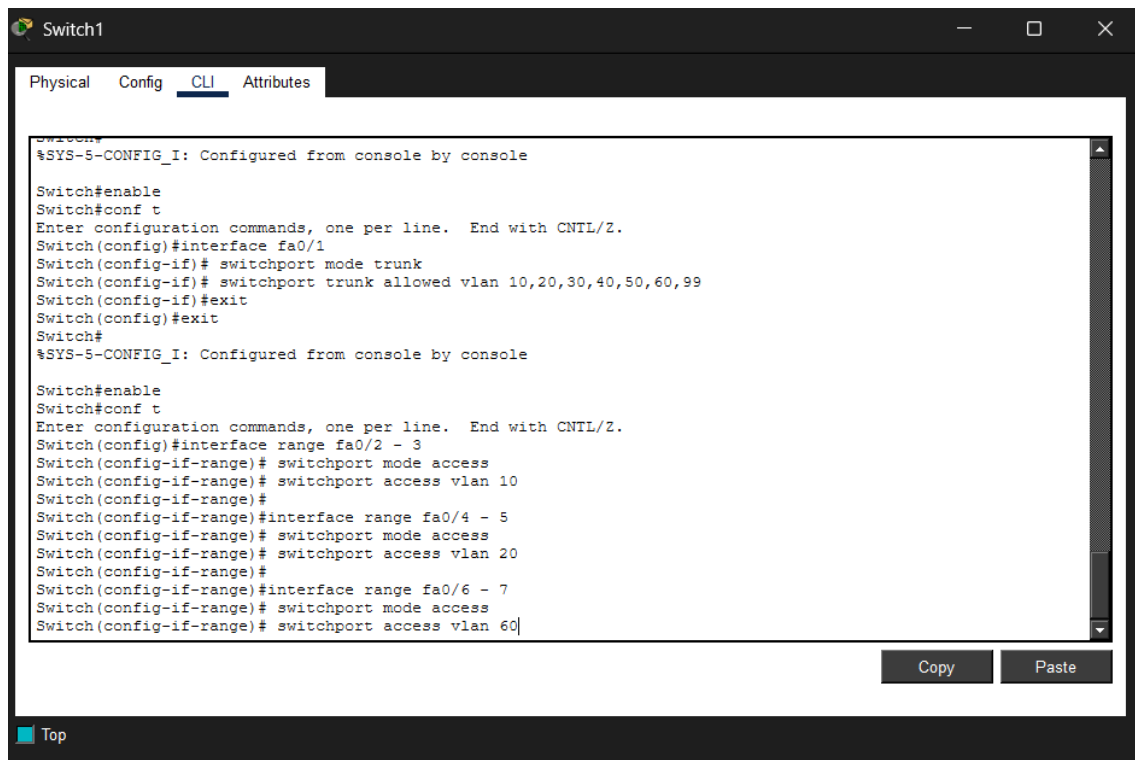
Luego en los switch de plantas.



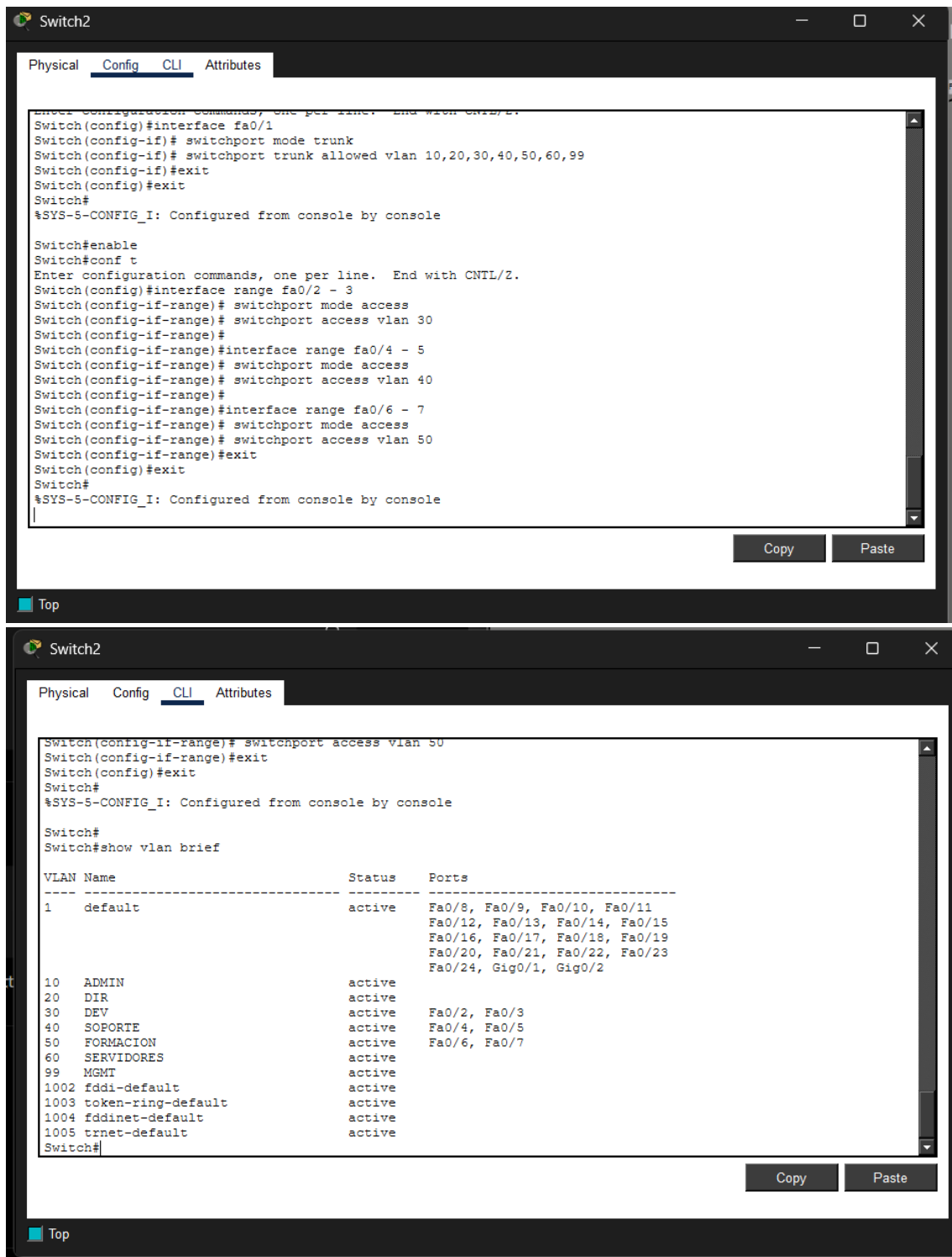


5. Configuro los puertos de acceso.

Primero en las vlan del switch 1



Switch 2



6. Configuro las DHCP en le switch de capa 3

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip dhcp excluded-address 192.168.10.1 192.168.10
Switch(config)#ip dhcp pool ADMIN
Switch(dhcp-config)# network 192.168.10.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.10.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool DIR
Switch(dhcp-config)# network 192.168.20.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.20.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool DEV
Switch(dhcp-config)# network 192.168.30.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.30.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool SOPORTE
Switch(dhcp-config)# network 192.168.40.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.40.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool FORMACION
Switch(dhcp-config)# network 192.168.50.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.50.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool SERVIDORES
Switch(dhcp-config)# network 192.168.60.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.60.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool MGMT
Switch(dhcp-config)# network 192.168.99.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.99.1|

```

7. Creo las Access List. En el switch de capa 3

```

Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#access-list 100 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
Switch(config)#access-list 100 permit ip any any
Switch(config)#
Switch(config)#interface vlan 50
Switch(config-if)# ip access-group 100 in
Switch(config-if)#
Switch(config-if)#access-list 101 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
Switch(config-if)#access-list 101 permit ip any any
Switch(config)#interface vlan 30
Switch(config-if)# ip access-group 101 in
Switch(config-if)#
Switch(config-if)#access-list 102 permit ip 192.168.99.0 0.0.0.255 any
Switch(config)#access-list 102 deny ip any any
Switch(config)#line vty 0 4
Switch(config-line)# access-class 102 in
Switch(config-line)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
|

```

Creo las Access list donde formación no puede acceder a dirección.

Desarrollo no puede acceder a administración

SOLO VLAN MGMT puede administrar.

Compruebo.

```

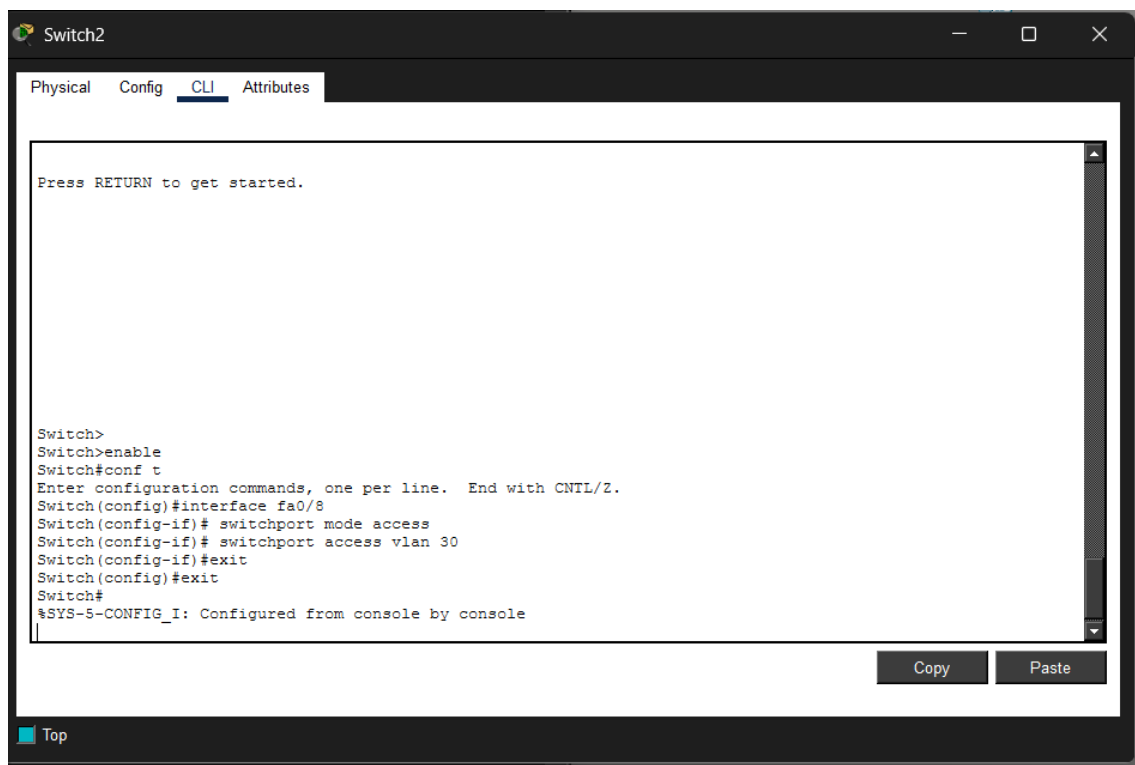
Switch#
Switch#show access-list
Extended IP access list 100
  10 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
  20 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
  30 permit ip any any
Extended IP access list 101
  10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
  20 permit ip any any
Extended IP access list 102
  10 permit ip 192.168.99.0 0.0.0.255 any
  20 deny ip any any

Switch#show run
Building configuration...

Current configuration : 3481 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
no profinet
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
!
--More-- |

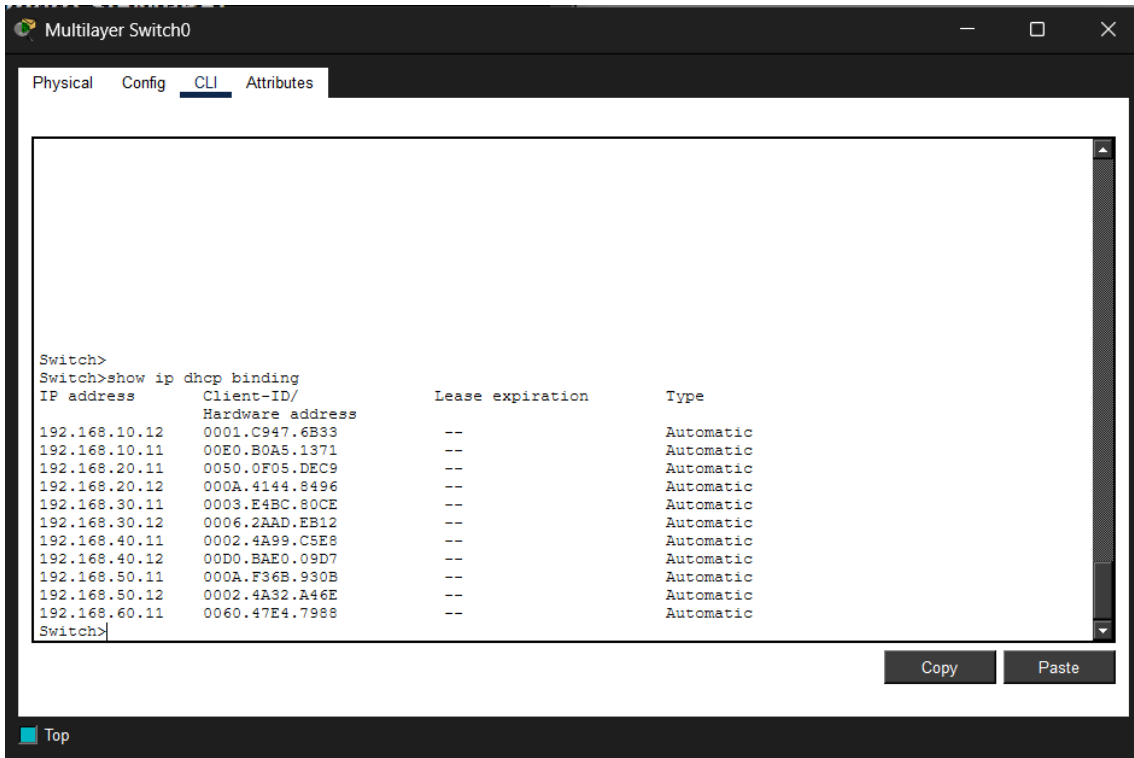
```

8. Configuro el wifi en el switch de planta primera.



9. Realizo pings para comprobar conectividad, accesos permitidos y DHCP funcionando.

--Compruebo DHCP en switch l3.

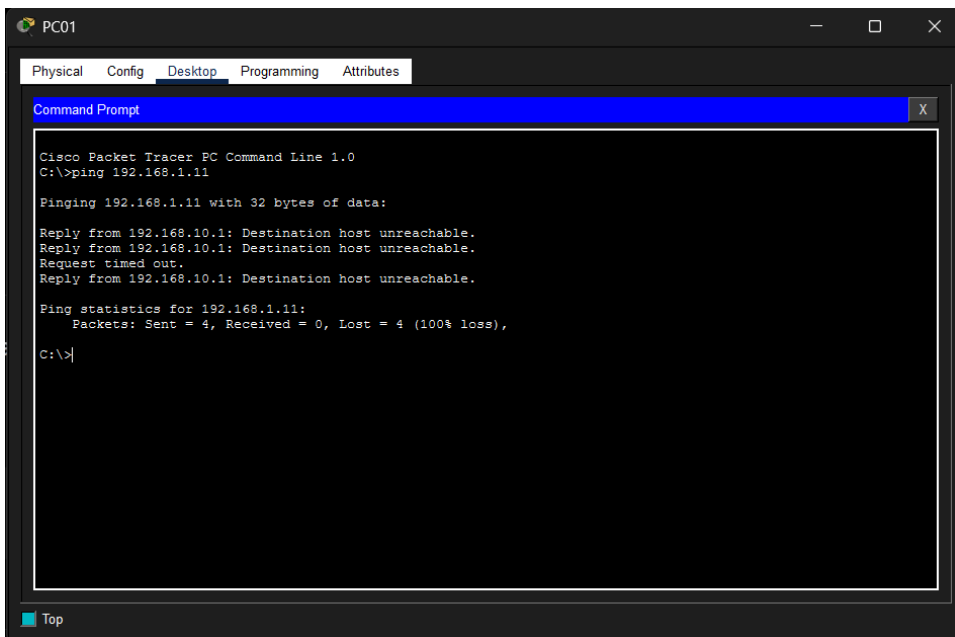


The screenshot shows the CLI of a Multilayer Switch0. The 'CLI' tab is selected. The command 'show ip dhcp binding' has been executed, displaying a table of DHCP bindings. The table has four columns: IP address, Client-ID/Hardware address, Lease expiration, and Type. All lease expiration values are '--' and all types are 'Automatic'.

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.10.12	0001.C947.6B33	--	Automatic
192.168.10.11	00E0.B0A5.1371	--	Automatic
192.168.20.11	0050.0F05.DEC9	--	Automatic
192.168.20.12	000A.4144.8496	--	Automatic
192.168.30.11	0003.E4BC.80CE	--	Automatic
192.168.30.12	0006.2AAD.EB12	--	Automatic
192.168.40.11	0002.4A99.C5E8	--	Automatic
192.168.40.12	00D0.BAE0.09D7	--	Automatic
192.168.50.11	000A.F36B.930B	--	Automatic
192.168.50.12	0002.4A32.A46E	--	Automatic
192.168.60.11	0060.47E4.7988	--	Automatic

--Hago ping dentro de una misma vlan

(Administracion)



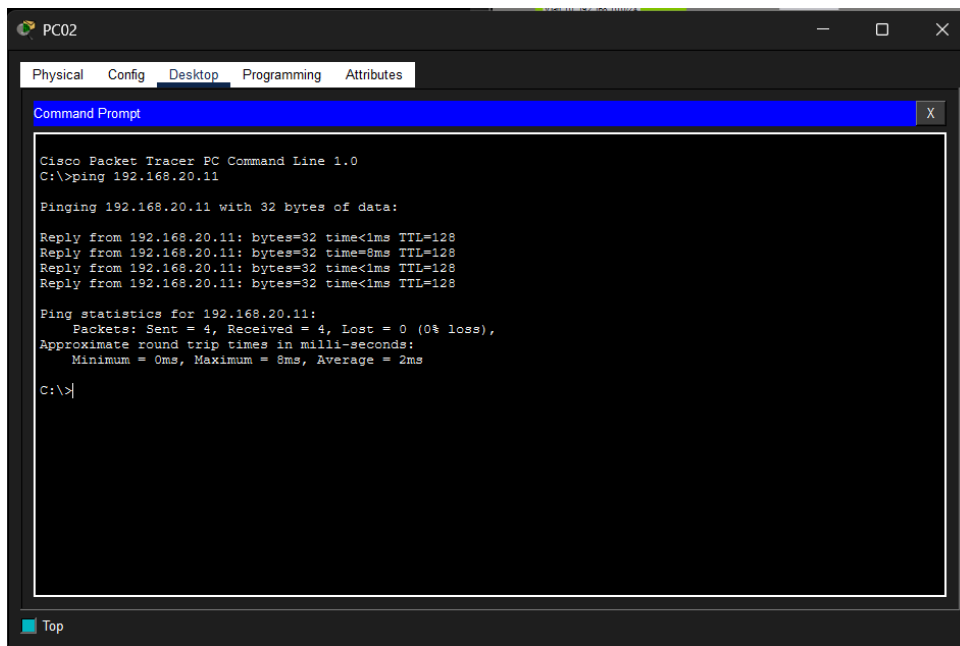
The screenshot shows the Command Prompt of PC01. The command 'ping 192.168.1.11' has been executed. The output shows four failed ping attempts, each with the message 'Destination host unreachable'. The ping statistics show 4 packets sent, 0 received, and 100% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

(Dirección)



```
PC02
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time=8ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>
```

El resto de ping funcionan perfectamente.

Fin del trabajo